

Scalable Column Subset Selection via Boolean Relaxation and Frank-Wolfe Method

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Abstract

The Column Subset Selection Problem (CSSP) is a major challenge in high-dimensional data analysis because of its NP-hard combinatorial nature. There is a need for approximation methods that are accurate but also fast enough to run on large datasets. Currently, most approaches either rely on “greedy” or “branch and bound” methods (Floudas, 2023). This thesis explores the theoretical link between the CSSP and the Minimum-Variance Portfolio Selection problem, which originated from the frameworks established by Markowitz (1952). We adapt the Boolean relaxation framework from Moka et al. (2025), which was originally used for optimizing financial portfolios, and apply it to column subset selection. We designed a new algorithm that efficiently finds a solution of CSSP, fixing the scalability issues that slow down traditional methods.

1 Introduction

The problem of lower-rank matrix approximation dates back to Eckart and Young (1936), who proved that a general solution exists when least-squares criterion of approximation is considered. The CSSP can be viewed as a constrained version of the Eckart-Young low-rank approximation problem. Unlike the general case, which allows for arbitrary basis vectors, CSSP requires the approximation to be formed strictly by selecting a subset of the original data columns. Mathematically, CSSP is a Mixed Integer Nonlinear Programming (MINLP) problem. We use binary variables (0 or 1) to decide if a column is included or not. As Burer and Letchford (2012) notes, non-convex MINLP problems are specifically challenging to solve because of their nature, often turning out to be NP-hard. The standard ways to solve this range from exact methods like Branch-and-Reduce by Ryoo and Sahinidis (1995) to faster Greedy algorithms by Ordozgoiti et al. (2019). However, both approaches have big flaws. Exact solvers (like Branch-and-Reduce) give the perfect answer, but they take exponential time, so they cannot be used on big datasets without sacrificing computational efficiency. On the other hand, Greedy algorithms are fast enough to run, but often lack regularization, which means they can overfit or be sensitive to noise (Ordozgoiti et al., 2019), and suffer when the subset size grows.

To solve this trade-off, this thesis looks at *continuous relaxation*. Instead of forcing our selection vector s to be exactly 0 or 1, we relax it to be continuous (between 0 and 1). This allows us to use gradient-based optimization, which is much faster (Moka et al., 2023). Specifically, we use the *Boolean relaxation framework* by Moka et al. (2025). Originally made for portfolio optimization, but we showed it works for CSSP. To solve this relaxed problem efficiently, we developed a novel adaptation of **Frank-Wolfe Homotopy** algorithm (Jaggi, 2013; Frank and Wolfe, 1956), which lets us smoothly move from a simplified convex optimisation to challenging MINLP ensuring computational scalability without significantly degrading the objective value.

FW-Homotopy, our newly designed algorithm performed significantly faster than its competitor Greedy, up to 5 to 6 times faster, while maintaining an acceptable objective value as p increased. Also, results from different datasets suggest that our FW-Homotopy is highly indifferent to the characteristic of

dataset like correlations. Rather, p and k turns out to be the most significant factor influencing its performance. This suggests FW-Homotopy's versatile usage potential across different datasets. However, FW-Homotopy was proven to be not the best choice for situations where p was small, and when sparse regime (where k/p ratio is small) was considered. Moreover, empirical results indicate that while FW-Homotopy offers significant computational speedups, the objective value remains comparable to the Greedy baseline, suggesting the relaxation is stable, but not necessarily superior in objective value for the tested regime.

This thesis makes following main contributions:

1. **Theoretical Reformulation:** We prove why Boolean Relaxation method used in portfolio selection could be used for CSSP.
2. **Algorithmic Innovation:** We developed **FW-Homotopy**, a novel adaptation of the Frank-Wolfe Homotopy algorithm specifically engineered to solve the continuous relaxation of the CSSP.
3. **Parameter Tuning:** We found that larger sample sizes m , or smaller step size α do not always mean better convergence, and identified the optimal range in terms of these parameters.
4. **Scalability:** Benchmarks against Greedy were conducted, and results showed a stability threshold at $k/p \approx 0.1$ for large p . We further developed to build a guideline for using FW-Homotopy.

The remainder of this thesis is organized as follows. Chapter 2 provides the formal definition of CSSP, and its reformulation and boolean relaxation, with proof. Chapter 3 introduces the proposed algorithm FW-Homotopy, the adaptation of the Frank-Wolfe Homotopy algorithm, and in Chapter 4 presents the results of numericals on various datasets.

2 Methodology of the Column Subset Selection Problem

2.1 Column Subset Selection Problem

We first introduce the notation. The selection of columns is represented by a binary vector $\mathbf{s} \in \{0, 1\}^p$. The square sub-matrix of A , $A_{[\mathbf{s}]} \in \mathbb{R}^{|\mathbf{s}| \times |\mathbf{s}|}$ is obtained by removing all columns and rows of A for which $s_j = 0, j = 1, \dots, p$, and $\|\cdot\|_F$ is a Frobenius norm, defined as $\|A\|_F = (\text{Tr}(A^\top A))^{1/2}$.

The Nyström approximation (Drineas and Mahoney, 2005; Williams and Seeger, 2001) is a widely adopted variant of the Column Subset Selection Problem (CSSP).

$$\min_{\mathbf{s} \in \{0,1\}^p: |\mathbf{s}| \leq k} \|X - P_{\mathbf{s}} X\|_F^2, \quad (1)$$

where the projection matrix is defined as

$$P_{\mathbf{s}} := X_{[:,\mathbf{s}]} (X_{[:,\mathbf{s}]}^\top X_{[:,\mathbf{s}]})^\dagger X_{[:,\mathbf{s}]}^\top,$$

where $X^\dagger$ denotes the Moore-Penrose pseudo-inverse of X , and $X \in \mathbb{R}^{m \times p}$ is a data matrix.

Objective function of (1) can be reformed into

$$\|X - P_{\mathbf{s}} X\|_F^2 = \|X\|_F^2 - \text{Tr}(X^\top P_{\mathbf{s}} X), \quad (2)$$

using $\|B\|_F^2 = \text{Tr}(B^\top B)$ and the idempotence of $P_{\mathbf{s}}$ ($P_{\mathbf{s}}^2 = P_{\mathbf{s}}$). Hence, (1) is equivalent to

$$\max_{\mathbf{s} \in \{0,1\}^p: |\mathbf{s}| \leq k} \text{Tr}(X^\top P_{\mathbf{s}} X). \quad (3)$$

since $\|X\|_F^2$ is a constant in terms of $\mathbf{s}$.

We now use a Rademacher vector $\boldsymbol{\xi} \in \{\pm 1\}^n$ with $\mathbb{E}[\boldsymbol{\xi}\boldsymbol{\xi}^\top] = I$, and by Hutchinson's identity (Hutchinson, 1989), $\text{Tr}(B) = \mathbb{E}_{\boldsymbol{\xi}}[\boldsymbol{\xi}^\top B \boldsymbol{\xi}]$ for any square B . Hence, (3) is equivalent to

$$\text{Tr}(X^\top P_{\mathbf{s}} X) = \mathbb{E}_{\boldsymbol{\xi}}[\boldsymbol{\xi}^\top X^\top P_{\mathbf{s}} X \boldsymbol{\xi}]. \quad (4)$$

Lastly, let

$$A := X^\top X \in \mathbb{R}^{n \times n}, \quad \mathbf{b}^{(\boldsymbol{\xi})} := X^\top X \boldsymbol{\xi} \in \mathbb{R}^n,$$

such that $A_{[\mathbf{s}]} = X_{[:,\mathbf{s}]}^\top X_{[:,\mathbf{s}]}$ and $\mathbf{b}_{[\mathbf{s}]}^{(\boldsymbol{\xi})} = X_{[:,\mathbf{s}]}^\top X \boldsymbol{\xi}$. Then, (4) becomes

$$\boldsymbol{\xi}^\top X^\top P_{\mathbf{s}} X \boldsymbol{\xi} = (\mathbf{b}_{[\mathbf{s}]}^{(\boldsymbol{\xi})})^\top (A_{[\mathbf{s}]})^\dagger \mathbf{b}_{[\mathbf{s}]}^{(\boldsymbol{\xi})},$$

and the CSSP objective can be expressed as the following form:

$$\max_{\mathbf{s} \in \{0,1\}^p: |\mathbf{s}| \leq k} \mathbb{E}_{\boldsymbol{\xi}}[(\mathbf{b}_{[\mathbf{s}]}^{(\boldsymbol{\xi})})^\top (A_{[\mathbf{s}]})^\dagger \mathbf{b}_{[\mathbf{s}]}^{(\boldsymbol{\xi})}]. \quad (5)$$

2.2 Boolean Relaxation

Since $\mathbf{b}_{[\mathbf{s}]}^{(\boldsymbol{\xi}_i)} = \mathbf{1} \odot \mathbf{b}_{[\mathbf{s}]}^{(\boldsymbol{\xi}_i)}$, (5) can be reformulated using the Hadamard product, into

$$\max_{\mathbf{s} \in \{0,1\}^p: |\mathbf{s}| \leq k} \mathbb{E}_{\boldsymbol{\xi}}[(\mathbf{1} \odot \mathbf{b}_{[\mathbf{s}]}^{(\boldsymbol{\xi}_i)})^\top (A_{[\mathbf{s}]})^{-1} (\mathbf{1} \odot \mathbf{b}_{[\mathbf{s}]}^{(\boldsymbol{\xi}_i)})]. \quad (6)$$

We propose the boolean relaxation of (6) as,

$$\max_{\mathbf{t} \in \mathcal{C}_k} g_\delta(\mathbf{t}) = \max_{\mathbf{t} \in \mathcal{C}_k} \mathbb{E}_{\boldsymbol{\xi}}[f_\delta(\mathbf{t}, \boldsymbol{\xi})], \quad (7)$$

where for each k , the constraint set $\mathcal{C}_k$ is a polytope defined as

$$\mathcal{C}_k = \{\mathbf{t} \in [0, 1]^p : \mathbf{t}^\top \mathbf{1} \leq k\}. \quad (8)$$

Additionally,

$$f_\delta(\mathbf{t}, \boldsymbol{\xi}) = [\mathbf{b}^{(\boldsymbol{\xi})} \odot \mathbf{t}]^\top (\tilde{A}_{\mathbf{t}})^{-1} [\mathbf{b}^{(\boldsymbol{\xi})} \odot \mathbf{t}], \quad (9)$$

and

$$\tilde{A}_{\mathbf{t}} = T_{\mathbf{t}} A T_{\mathbf{t}} + \delta(I - T_{\mathbf{t}}^2). \quad (10)$$

This was inspired by the idea that inner subproblem has a same form of minimum variance portfolio problem, specifically scaled problems (Moka et al., 2025).

2.3 Theoretical Properties

Theorem (1) below shows why (7) is a boolean relaxation of the original problem (5).

Theorem 1 (Modified from Theorems 2 & 3 of Moka et al. (2025)). *The following hold:*

- (i) *Continuity: The objective function $g_\delta(\mathbf{t})$ in (7) is continuous on $[0, 1]^p$.*

(ii) *Corner point agreement: For every binary vector $\mathbf{s} \in \{0, 1\}^p$ (i.e., a corner point on the hypercube $[0, 1]^p$),*

$$g_\delta(\mathbf{s}) = \mathbb{E}_{\boldsymbol{\xi}} \left[[\mathbf{b}^{(\boldsymbol{\xi}_i)} \odot \mathbf{s}]^T A_{\mathbf{s}}^{-1} [\mathbf{b}^{(\boldsymbol{\xi}_i)} \odot \mathbf{s}] \right], \quad \text{for all } \delta > 0.$$

(iii) *For every fixed $\mathbf{t} \in (0, 1)^p$, $g_\delta(\mathbf{t})$ is monotonically decreasing in $\delta > 0$.*

(iv) *For any $k = 1, \dots, p$ and $\delta > 0$,*

$$\max_{\mathbf{t} \in \mathcal{C}_k} g_\delta(\mathbf{t}) = \max_{\mathbf{t} \in \mathcal{S}_k} g_\delta(\mathbf{t}),$$

here, the simplex $\mathcal{S}_k = \{\mathbf{t} \in [0, 1]^p : \mathbf{t}^\top \mathbf{1} = k\}$ corresponds to the polytope $\mathcal{C}_k$ given in (8).

(v) *Let η_1 be the largest eigenvalue of A . Then, $g_\delta(\mathbf{t})$ strictly convex over $[0, 1]^p$ for $\delta \geq \eta_1$.*

Theorem 2. *The following hold:*

(i) *Partial gradient of $f_\delta(\mathbf{t}, \boldsymbol{\xi})$ with respect to $\mathbf{t}$ is given as,*

$$\nabla_{\mathbf{t}} f_\delta(\mathbf{t}, \boldsymbol{\xi}) = 2\delta \frac{(\Pi_{\mathbf{t}}^{-1} [\mathbf{b}^{(\boldsymbol{\xi})} \odot \mathbf{1}])^2}{\mathbf{t}^3}, \quad \mathbf{t} \in (0, 1)^p.$$

(ii) *Gradient of $g_\delta(\mathbf{t})$ is given as,*

$$\nabla g_\delta(\mathbf{t}) = \frac{1}{2^n} \sum_{i=1}^{2^n} 2\delta \frac{(\Pi_{\mathbf{t}}^{-1} [\mathbf{b}^{(\boldsymbol{\xi}_i)} \odot \mathbf{1}])^2}{\mathbf{t}^3}.$$

where

$$\Pi_{\mathbf{t}} = A + \delta D_{\mathbf{t}} \quad \text{and} \quad D_{\mathbf{t}} = T_{\mathbf{t}}^{-2} - I.$$

All the proofs can be found in next section.

2.4 Proof.

We prove that our boolean relaxation (7) satisfies all the properties of (1).

From Moka et al. (2025), we already know that there exists a valid boolean relaxation for a original problem of form

$$\min_{\mathbf{s} \in \{0, 1\}^p} -\mathbf{1}^\top \Sigma_{[\mathbf{s}]}^{-1} \mathbf{1}.$$

Now, let $\Sigma_{[\mathbf{s}]}^{-1} = \text{Diag}(\mathbf{b}_{[\mathbf{s}]}^{(\boldsymbol{\xi}_i)}) A_{[\mathbf{s}]}^{-1} \text{Diag}(\mathbf{b}_{[\mathbf{s}]}^{(\boldsymbol{\xi}_i)})$, then original problem is equivalent to,

$$\begin{aligned} \min_{\mathbf{s} \in \{0, 1\}^p} -\mathbf{1}^\top \Sigma_{[\mathbf{s}]}^{-1} \mathbf{1} &= \max_{\mathbf{s} \in \{0, 1\}^p} \mathbf{1}^\top \Sigma_{[\mathbf{s}]}^{-1} \mathbf{1} \\ &= \max_{\mathbf{s} \in \{0, 1\}^p} \mathbf{1}^\top \text{Diag}(\mathbf{b}_{[\mathbf{s}]}^{(\boldsymbol{\xi}_i)}) A_{[\mathbf{s}]}^{-1} \text{Diag}(\mathbf{b}_{[\mathbf{s}]}^{(\boldsymbol{\xi}_i)}) \mathbf{1} \\ &= \max_{\mathbf{s} \in \{0, 1\}^p} (\mathbf{1} \odot \mathbf{b}_{[\mathbf{s}]}^{(\boldsymbol{\xi}_i)})^\top A_{[\mathbf{s}]}^{-1} (\mathbf{1} \odot \mathbf{b}_{[\mathbf{s}]}^{(\boldsymbol{\xi}_i)}), \end{aligned}$$

using $\text{Diag}(\mathbf{b}_{[\mathbf{s}]}^{(\boldsymbol{\xi}_i)}) \mathbf{1} = \mathbf{1} \odot \mathbf{b}_{[\mathbf{s}]}^{(\boldsymbol{\xi}_i)}$.

Hence, inner term $(\mathbf{1} \odot \mathbf{b}_{[\mathbf{s}]}^{(\boldsymbol{\xi}_i)})^\top (A_{[\mathbf{s}]}^{-1}) (\mathbf{1} \odot \mathbf{b}_{[\mathbf{s}]}^{(\boldsymbol{\xi}_i)})$ is a special form of minimum-variance portfolio selection problem. Additionally, minimum-variance portfolio problem

$$\max_{\mathbf{s} \in \{0, 1\}^p : |\mathbf{s}| \leq k} (\mathbf{1} \odot \mathbf{b}_{[\mathbf{s}]}^{(\boldsymbol{\xi}_i)})^\top (A_{[\mathbf{s}]}^{-1}) (\mathbf{1} \odot \mathbf{b}_{[\mathbf{s}]}^{(\boldsymbol{\xi}_i)}) \quad (11)$$

can be relaxed into

$$\max_{\mathbf{t} \in \mathcal{C}_k} f_\delta(\mathbf{t}) \quad (12)$$

and this relaxation satisfies all the properties of (1) (Moka et al., 2025).

2.4.1 (i)

Continuity of $g_\delta(\mathbf{t})$ is from the fact that each $f_\delta(\mathbf{t}, \boldsymbol{\xi})$ is continuous on $[0, 1]^p$, hence summation of 2^n continuous functions will also be continuous in the same hypercube.

2.4.2 (ii)

For any binary vector $\mathbf{s} \in \{0, 1\}^p$,

$$(\tilde{A}_\mathbf{s})_{[\mathbf{s}]} = (T_\mathbf{s} A T_\mathbf{s})_{[\mathbf{s}]} = A_{[\mathbf{s}]} \quad \text{and} \quad (\tilde{A}_\mathbf{s})_{[\mathbf{1}-\mathbf{s}]} = \delta I.$$

$$\text{Hence, } f_\delta(\mathbf{s}) = [\mathbf{b}^{(\boldsymbol{\xi})} \odot \mathbf{s}]^T A_{[\mathbf{s}]}^{-1} [\mathbf{b}^{(\boldsymbol{\xi})} \odot \mathbf{s}].$$

As a consequence, $g_\delta(\mathbf{t})$ is equal to

$$\begin{aligned} g_\delta(\mathbf{s}) &= \mathbb{E}_{\boldsymbol{\xi}} \left[f_\delta(\mathbf{t}, \boldsymbol{\xi}) \right] \\ &= \frac{1}{2^n} \sum_{i=1}^{2^n} f_\delta(\mathbf{s}, \boldsymbol{\xi}_i) = \sum_{i=1}^{2^n} [\mathbf{b}^{(\boldsymbol{\xi}_i)} \odot \mathbf{s}]^T A_{[\mathbf{s}]}^{-1} [\mathbf{b}^{(\boldsymbol{\xi}_i)} \odot \mathbf{s}] = \mathbb{E}_{\boldsymbol{\xi}} \left[[\mathbf{b}^{(\boldsymbol{\xi}_i)} \odot \mathbf{s}]^T A_{[\mathbf{s}]}^{-1} [\mathbf{b}^{(\boldsymbol{\xi}_i)} \odot \mathbf{s}] \right], \end{aligned}$$

, since $\mathbb{P}(\boldsymbol{\xi} = \boldsymbol{\xi}_i) = \frac{1}{2^n}$. Hence, we showed that at every corner point on the hypercube, our relaxation's cost function is equivalent to original problem.

2.4.3 (iii)

Lemma 1. (Adaptation of lemma 2 from Moka et al. (2025))

For every interior point $\mathbf{t} \in (0, 1)^p$, our objective function $f_\delta(\mathbf{t})$ can be expressed as

$$f_\delta(\mathbf{t}, \boldsymbol{\xi}) = [\mathbf{b}^{(\boldsymbol{\xi})} \odot \mathbf{1}]^T \Pi_\mathbf{t}^{-1} [\mathbf{b}^{(\boldsymbol{\xi})} \odot \mathbf{1}],$$

where

$$\Pi_\mathbf{t} = A + \delta D_\mathbf{t} \quad \text{and} \quad D_\mathbf{t} = T_\mathbf{t}^{-2} - I.$$

Proof. Since $\mathbf{t} = \mathbf{1}^\top T_\mathbf{t}$, $\mathbf{b}^{(\boldsymbol{\xi})} = (\mathbf{b}^{(\boldsymbol{\xi})} \odot \mathbf{1}^\top T_\mathbf{t}) = (\mathbf{b}^{(\boldsymbol{\xi})} \odot \mathbf{1}) T_\mathbf{t}$ holds. Hence,

$$\begin{aligned} f_\delta(\mathbf{t}, \boldsymbol{\xi}) &= (\mathbf{b}^{(\boldsymbol{\xi})} \odot \mathbf{1})^\top T_\mathbf{t} \tilde{A}_\mathbf{t}^{-1} T_\mathbf{t} (\mathbf{b}^{(\boldsymbol{\xi})} \odot \mathbf{1}) \\ &= (\mathbf{b}^{(\boldsymbol{\xi})} \odot \mathbf{1})^\top (T_\mathbf{t}^{-1} \tilde{A}_\mathbf{t} T_\mathbf{t}^{-1})^{-1} (\mathbf{b}^{(\boldsymbol{\xi})} \odot \mathbf{1}) \\ &= (\mathbf{b}^{(\boldsymbol{\xi})} \odot \mathbf{1})^\top (A + \delta(T_\mathbf{t}^{-2} - I))^{-1} (\mathbf{b}^{(\boldsymbol{\xi})} \odot \mathbf{1}) \end{aligned}$$

□

Using above result, we get

$$\begin{aligned}
\frac{d}{d\delta} f_\delta(\mathbf{t}, \boldsymbol{\xi}) &= [\mathbf{b}^{(\boldsymbol{\xi})} \odot \mathbf{1}]^\top \left(\frac{d}{d\delta} \Pi_{\mathbf{t}}^{-1} \right) [\mathbf{b}^{(\boldsymbol{\xi})} \odot \mathbf{1}] \\
&= -[\mathbf{b}^{(\boldsymbol{\xi})} \odot \mathbf{1}]^\top \Pi_{\mathbf{t}}^{-1} \left(\frac{d}{d\delta} \Pi_{\mathbf{t}} \right) \Pi_{\mathbf{t}}^{-1} [\mathbf{b}^{(\boldsymbol{\xi})} \odot \mathbf{1}] \\
&= -\mathbf{u}^\top \left(\frac{d}{d\delta} \Pi_{\mathbf{t}} \right) \mathbf{u} \\
&= \mathbf{u}^\top (I - T_{\mathbf{t}}^{-2}) \mathbf{u},
\end{aligned}$$

with $\mathbf{u} = \Pi_{\mathbf{t}}^{-1}[\mathbf{b}^{(\boldsymbol{\xi})} \odot \mathbf{1}] \neq \mathbf{0}$. Since $\mathbf{t}$ is in unit interval, the matrix $I - T_{\mathbf{t}}^{-2}$ in the quadratic form is negative definite. Hence, $f_\delta(\mathbf{t}, \boldsymbol{\xi})$ is monotonically *decreasing* in $\delta > 0$, and naturally, $g_\delta(\mathbf{t})$ is also monotonically decreasing, as $ig_\delta(\mathbf{t})$ is a summation of monotonic decreasing functions.

2.4.4 (iv)

We can find the gradient of our relaxed objective using (1), with a minor adjustment.

For every interior point $\mathbf{t} \in (0, 1)^p$, our objective function $f_\delta(\mathbf{t})$ can be expressed as

$$f_\delta(\mathbf{t}, \boldsymbol{\xi}) = [\mathbf{b}^{(\boldsymbol{\xi})} \odot \mathbf{1}]^T \Pi_{\mathbf{t}}^{-1} [\mathbf{b}^{(\boldsymbol{\xi})} \odot \mathbf{1}].$$

Thus, for each $j = 1, \dots, p$,

$$\begin{aligned}
\frac{\partial f_\delta(\mathbf{t}, \boldsymbol{\xi})}{\partial t_j} &= [\mathbf{b}^{(\boldsymbol{\xi})} \odot \mathbf{1}]^T \frac{\partial \Pi_{\mathbf{t}}^{-1}}{\partial t_j} [\mathbf{b}^{(\boldsymbol{\xi})} \odot \mathbf{1}] \\
&= -[\mathbf{b}^{(\boldsymbol{\xi})} \odot \mathbf{1}]^T \Pi_{\mathbf{t}}^{-1} \frac{\partial \Pi_{\mathbf{t}}}{\partial t_j} \Pi_{\mathbf{t}}^{-1} [\mathbf{b}^{(\boldsymbol{\xi})} \odot \mathbf{1}].
\end{aligned}$$

Additionally, since A doesn't depend on $\mathbf{t}$ from our definition $\Pi_{\mathbf{t}} = A + \delta D_{\mathbf{t}}$,

$$\frac{\partial \Pi_{\mathbf{t}}}{\partial t_j} = \delta \frac{\partial D_{\mathbf{t}}}{\partial t_j} = -\frac{2\delta}{t_j^3} \mathbf{e}_j \mathbf{e}_j^\top.$$

Thus,

$$\nabla f_\delta(\mathbf{t}, \boldsymbol{\xi}) = 2\delta \frac{(\Pi_{\mathbf{t}}^{-1}[\mathbf{b}^{(\boldsymbol{\xi})} \odot \mathbf{1}])^2}{t^3}, \quad \mathbf{t} \in (0, 1)^p.$$

For interior points,

$$\begin{aligned}
\nabla g_\delta(\mathbf{t}) &= \nabla \left(\frac{1}{2^n} \sum_{i=1}^{2^n} f_\delta(\mathbf{t}, \boldsymbol{\xi}_i) \right) = \frac{1}{2^n} \sum_{i=1}^{2^n} \nabla f_\delta(\mathbf{t}, \boldsymbol{\xi}_i) \\
&= \frac{1}{2^n} \sum_{i=1}^{2^n} 2\delta \frac{(\Pi_{\mathbf{t}}^{-1}[\mathbf{b}^{(\boldsymbol{\xi}_i)} \odot \mathbf{1}])^2}{t^3}.
\end{aligned} \tag{13}$$

Note that, all the elements of gradient $\nabla g_\delta(\mathbf{t}, \boldsymbol{\xi})$ are non-negative since each $\nabla f_\delta(\mathbf{t}, \boldsymbol{\xi})$ are non-negative. Hence, objective function $g_\delta(\mathbf{t}, \boldsymbol{\xi})$ is non-decreasing along t_j . Hence, simplex S_k contains an optimal solution of boolean relaxation.

2.4.5 (v)

Let η_1, η_p be the largest and smallest eigenvalues of A , respectively. Then, for each $f_\delta(\mathbf{t}, \boldsymbol{\xi})$, below holds, by **Theorem 3** of Moka et al. (2025).

- (i) $f_\delta(\mathbf{t}, \boldsymbol{\xi})$ is strictly convex over $[0, 1]^p$ for $\delta \geq \eta_1$
- (ii) For any $\epsilon \in (0, 1)$, $f_\delta(\mathbf{t}, \boldsymbol{\xi})$ is strictly concave over $[\epsilon, 1]^p$ for $\delta \leq 3\eta_p\epsilon^2/(1 + 3\epsilon^2)$.

Since our A is a constant in terms of $\boldsymbol{\xi}$ and $\mathbf{t}$, this is true for each 2^n terms of $f_\delta(\mathbf{t}, \boldsymbol{\xi}_i)$. Moreover, finite sum of strictly convex / concave functions will also be a strictly convex / concave function respectively, we know that,

- (i) $g_\delta(\mathbf{t})$ is strictly convex over $[0, 1]^p$ for $\delta \geq \eta_1$
- (ii) For any $\epsilon \in (0, 1)$, $g_\delta(\mathbf{t})$ is strictly concave over $[\epsilon, 1]^p$ for $\delta \leq 3\eta_p\epsilon^2/(1 + 3\epsilon^2)$.

We also derive the Hessian.

Lemma 2. (i) Hessian of $f_\delta(\mathbf{t}, \boldsymbol{\xi})$ for every interior point $\mathbf{t} \in (0, 1)^p$ is given by

$$H_\delta(\mathbf{t}, \boldsymbol{\xi}) = 2\delta \text{Diag}\left(\frac{\mathbf{u}_{\mathbf{t}, \boldsymbol{\xi}}}{t^3}\right) [4\delta \Pi_{\mathbf{t}}^{-1} - 3 \text{Diag}(\mathbf{t}^2)] \text{Diag}\left(\frac{\mathbf{u}_{\mathbf{t}, \boldsymbol{\xi}}}{t^3}\right),$$

$$\text{where } \Pi_{\mathbf{t}} = A + \delta D_{\mathbf{t}} \quad \text{and} \quad D_{\mathbf{t}} = T_{\mathbf{t}}^{-2} - I \quad \text{and} \quad \mathbf{u}_{\mathbf{t}, \boldsymbol{\xi}} = \Pi_{\mathbf{t}}^{-1}[\mathbf{b}^{(\boldsymbol{\xi})} \odot \mathbf{1}] = \Pi_{\mathbf{t}}^{-1}\mathbf{b}^{(\boldsymbol{\xi})}.$$

(ii) Hessian of $g_\delta(\mathbf{t})$ as,

$$H_{g, \delta}(\mathbf{t}) = \frac{1}{2^n} \sum_{i=1}^{2^n} H_{f, \delta}(\mathbf{t}, \boldsymbol{\xi}_i)$$

Proof. Using

$$\nabla f_\delta(\mathbf{t}, \boldsymbol{\xi}) = 2\delta \frac{(\Pi_{\mathbf{t}}^{-1}[\mathbf{b}^{(\boldsymbol{\xi})} \odot \mathbf{1}])^2}{t^3}, \quad \mathbf{t} \in (0, 1)^p.$$

, we can express derivative to t_j as below,

$$\begin{aligned} \frac{\partial}{\partial t_j} \nabla f_\delta(\mathbf{t}, \boldsymbol{\xi}) &= \frac{-6\delta}{t_j^4} \mathbf{e}_j \mathbf{e}_j^\top (\mathbf{u}_{\mathbf{t}, \boldsymbol{\xi}} \odot \mathbf{u}_{\mathbf{t}, \boldsymbol{\xi}}) + 2\delta \left(\frac{1}{t^3}\right) \odot \frac{\partial}{\partial t_j} (\mathbf{u}_{\mathbf{t}, \boldsymbol{\xi}} \odot \mathbf{u}_{\mathbf{t}, \boldsymbol{\xi}}) \\ &= \frac{-6\delta}{t_j^4} \mathbf{e}_j \mathbf{e}_j^\top (\mathbf{u}_{\mathbf{t}, \boldsymbol{\xi}} \odot \mathbf{u}_{\mathbf{t}, \boldsymbol{\xi}}) + 4\delta \left(\frac{\mathbf{u}_{\mathbf{t}, \boldsymbol{\xi}}}{t^3}\right) \odot \frac{\partial \mathbf{u}_{\mathbf{t}, \boldsymbol{\xi}}}{\partial t_j} \\ &= \frac{-6\delta}{t_j^4} \mathbf{e}_j \mathbf{e}_j^\top (\mathbf{u}_{\mathbf{t}, \boldsymbol{\xi}} \odot \mathbf{u}_{\mathbf{t}, \boldsymbol{\xi}}) - 4\delta \left(\frac{\mathbf{u}_{\mathbf{t}, \boldsymbol{\xi}}}{t^3}\right) \odot [\Pi_{\mathbf{t}}^{-1} \frac{\partial \Pi_{\mathbf{t}}^{-1}}{\partial t_j} \Pi_{\mathbf{t}}^{-1} \mathbf{b}^{(\boldsymbol{\xi})}] \\ &= \frac{-6\delta}{t_j^4} \mathbf{e}_j \mathbf{e}_j^\top (\mathbf{u}_{\mathbf{t}, \boldsymbol{\xi}} \odot \mathbf{u}_{\mathbf{t}, \boldsymbol{\xi}}) + 4\delta \left(\frac{\mathbf{u}_{\mathbf{t}, \boldsymbol{\xi}}}{t^3}\right) \odot [\Pi_{\mathbf{t}}^{-1} \frac{2\delta}{t_j^3} \mathbf{e}_j \mathbf{e}_j^\top \Pi_{\mathbf{t}}^{-1} \mathbf{b}^{(\boldsymbol{\xi})}] \end{aligned}$$

$$\text{using } \frac{\partial \Pi_{\mathbf{t}}}{\partial t_j} = \delta \frac{\partial D_{\mathbf{t}}}{\partial t_j} = -\frac{2\delta}{t_j^3} \mathbf{e}_j \mathbf{e}_j^\top.$$

Further expanding, above reformulates to

$$\frac{\partial}{\partial t_j} \nabla f_\delta(\mathbf{t}, \boldsymbol{\xi}) = \frac{-6\delta}{t_j^4} \mathbf{e}_j \mathbf{e}_j^\top (\mathbf{u}_{\mathbf{t}, \boldsymbol{\xi}} \odot \mathbf{u}_{\mathbf{t}, \boldsymbol{\xi}}) + \frac{8\delta^2}{t_j^3} \left(\frac{\mathbf{u}_{\mathbf{t}, \boldsymbol{\xi}}}{t^3}\right) \odot [\Pi_{\mathbf{t}}^{-1} \mathbf{e}_j \mathbf{e}_j^\top \mathbf{u}_{\mathbf{t}, \boldsymbol{\xi}}].$$

Hence, Hessian of $f_\delta(\mathbf{t}, \boldsymbol{\xi})$ is given by

$$\begin{aligned} H_{f, \delta}(\mathbf{t}, \boldsymbol{\xi}) &= -6\delta [\text{Diag}\left(\frac{\mathbf{u}_{\mathbf{t}, \boldsymbol{\xi}}}{t^3}\right)]^2 T_{\mathbf{t}}^2 + 8\delta^2 \text{Diag}\left(\frac{\mathbf{u}_{\mathbf{t}, \boldsymbol{\xi}}}{t^3}\right) \Pi_{\mathbf{t}}^{-1} \text{Diag}\left(\frac{\mathbf{u}_{\mathbf{t}, \boldsymbol{\xi}}}{t^3}\right) \\ &= 2\delta \text{Diag}\left(\frac{\mathbf{u}_{\mathbf{t}, \boldsymbol{\xi}}}{t^3}\right) [4\delta \Pi_{\mathbf{t}}^{-1} - 3 \text{Diag}(\mathbf{t}^2)] \text{Diag}\left(\frac{\mathbf{u}_{\mathbf{t}, \boldsymbol{\xi}}}{t^3}\right). \end{aligned}$$

Hence, we can also get Hessian of $g_\delta(\mathbf{t})$ as,

$$H_{g,\delta}(\mathbf{t}) = \frac{1}{2^n} \sum_{i=1}^{2^n} H_{f,\delta}(\mathbf{t}, \boldsymbol{\xi})$$

□

3 Implementation

For convenience, we reformulate original maximization problem (6) and (7) into a minimization problem below.

$$\begin{aligned} & \max_{\mathbf{s} \in \{0,1\}^p: |\mathbf{s}| \leq k} \mathbb{E}_{\boldsymbol{\xi}} \left[(\mathbf{1} \odot \mathbf{b}_{[\mathbf{s}]}^{(\boldsymbol{\xi}_i)})^\top (A_{[\mathbf{s}]})^{-1} (\mathbf{1} \odot \mathbf{b}_{[\mathbf{s}]}^{(\boldsymbol{\xi}_i)}) \right] \\ &= \min_{\mathbf{s} \in \{0,1\}^p: |\mathbf{s}| \leq k} -\mathbb{E}_{\boldsymbol{\xi}} \left[(\mathbf{1} \odot \mathbf{b}_{[\mathbf{s}]}^{(\boldsymbol{\xi}_i)})^\top (A_{[\mathbf{s}]})^{-1} (\mathbf{1} \odot \mathbf{b}_{[\mathbf{s}]}^{(\boldsymbol{\xi}_i)}) \right] \\ &= \min_{\mathbf{s} \in \{0,1\}^p: |\mathbf{s}| \leq k} \mathbb{E}_{\boldsymbol{\xi}} \left[-(\mathbf{1} \odot \mathbf{b}_{[\mathbf{s}]}^{(\boldsymbol{\xi}_i)})^\top (A_{[\mathbf{s}]})^{-1} (\mathbf{1} \odot \mathbf{b}_{[\mathbf{s}]}^{(\boldsymbol{\xi}_i)}) \right]. \end{aligned} \quad (14)$$

The boolean relaxation of (14) is

$$\min_{\mathbf{t} \in \mathcal{C}} \Phi_\delta(\mathbf{t}) = \min_{\mathbf{t} \in \mathcal{C}} \mathbb{E}_{\boldsymbol{\xi}} \left[h_\delta(\mathbf{t}, \boldsymbol{\xi}) \right] \quad (15)$$

, where $\Phi_\delta(\mathbf{t}) = \mathbb{E}_{\boldsymbol{\xi}} \left[h_\delta(\mathbf{t}, \boldsymbol{\xi}) \right]$, $h_\delta(\mathbf{t}, \boldsymbol{\xi}) = -f_\delta(\mathbf{t}, \boldsymbol{\xi}) = -[b^{(\boldsymbol{\xi})} \odot \mathbf{t}]^\top (\tilde{A}_t)^{-1} [b^{(\boldsymbol{\xi})} \odot \mathbf{t}]$, and $\tilde{A}_t = T_t A T_t + \delta(I - T_t^2)$.

Finally, gradient will be

$$\nabla h_\delta(\mathbf{t}, \boldsymbol{\xi}) = -\nabla f_\delta(\mathbf{t}, \boldsymbol{\xi}), \nabla \Phi_\delta(\mathbf{t}) = -\nabla g_\delta(\mathbf{t}).$$

Note that, both $\Phi_\delta(\mathbf{t})$ and $h_\delta(\mathbf{t}, \boldsymbol{\xi})$ are well-defined since $g_\delta(\mathbf{t})$ and $f_\delta(\mathbf{t}, \boldsymbol{\xi})$ are well-defined. Additionally, since $\nabla \Phi_\delta(\mathbf{t}) = -\nabla g_\delta(\mathbf{t})$, we know that

(i) $\Phi_\delta(\mathbf{t})$ is **strictly concave** over $[0, 1]^p$ for $\delta \geq \eta_1$

(ii) For any $\epsilon \in (0, 1)$, $\Phi_\delta(\mathbf{t})$ is **strictly convex** over $[\epsilon, 1]^p$ for $\delta \leq 3\eta_p \epsilon^2 / (1 + 3\epsilon^2)$.

Since $\Phi_\delta(\mathbf{t})$ is **strictly convex** for a small value of $\delta_0 = 3\eta_p \epsilon^2 / (1 + 3\epsilon^2)$, we start with a small value of δ to provide a stable initialisation. That is, the set up above for δ_0 guarantees any initialization leading to a unique global solution.

Building upon these theoretical foundation, algorithm FW-Homotopy iterates n times. For each epoch, it creates m random Rademacher vector samples, which will be used to find a *estimate* of gradient $\nabla \Phi_\delta(\mathbf{t})$. A selection vector $\mathbf{s} \in \{0, 1\}^p$, is chosen in a way that it denotes the k smallest components of $\nabla \Phi_\delta(\mathbf{t})$. That is, if $\nabla \Phi_\delta(\mathbf{t}) = (1 \ 2 \ 3 \ 4)^\top$ and $k = 2$, then $\mathbf{s} = (1 \ 1 \ 0 \ 0)^\top$. This step decides the direction of our step.

Next we take a weighted step toward the above direction, and restart the epoch.

Algorithm 1 FW-Homotopy(A, k, α, n, m) Moka et al. (2025)

```
1:  $n \leftarrow \max(20k, 100)$  for unspecified  $n$ .
2: Compute the largest and smallest eigenvalues  $\eta_1$  and  $\eta_p$  of  $A$ 
3:  $\varepsilon \leftarrow 0.1(k/p)$ 
4:  $\delta_0 \leftarrow 3\eta_p\varepsilon^2/(1 + 3\varepsilon^2)$ 
5:  $r \leftarrow (\eta_1/\delta_0)^{1/(n-1)}$ 
6:  $\mathbf{t} \leftarrow (k/p)\mathbf{1}$ 
7:  $\ell \leftarrow 1$ 
8: repeat
9:    $\delta \leftarrow \delta_0 r^\ell$ 
10:   Create  $m$  random Rademacher vectors  $\xi_i$  for  $i = 1, \dots, m$ 
11:   Estimate the gradient  $\nabla\Phi_\delta(\mathbf{t})$ , using  $m$  of  $\nabla h_\delta(\mathbf{t}, \xi_i)$  (2)
12:   Let  $\mathbf{s} \in \{0, 1\}^p$  have ones at the positions of the  $k$  smallest components of  $\nabla\Phi_\delta(\mathbf{t})$ 
13:    $\mathbf{t} \leftarrow (1 - \alpha)\mathbf{t} + \alpha\mathbf{s}$ 
14:    $\ell \leftarrow \ell + 1$ 
15: until  $\ell > n$ 
16: return  $\mathbf{s}$ 
```

3.1 Competitor - Greedy Algorithm

For comparison, we implemented a Greedy algorithm using forward selection (Kroese et al., 2019). The algorithm initializes with an empty set and iterates k times. For each iteration, it scans all remaining unselected columns and greedily adds a column that improves objective value the most.

We also reformulated (3)’s objective function can be reformulated as,

$$\begin{aligned} \text{Tr}(X^\top P_s X) &= \text{Tr}(X^\top X_{[:,s]} (A_{[s]})^\dagger X_{[:,s]}^\top X) \\ &= \text{Tr}((X_{[:,s]}^\top X_{[:,s]})^\dagger (X_{[:,s]}^\top X) (X^\top X_{[:,s]})) \\ &= \text{Tr}((A_{[s]})^{-1} X_{[:,s]}^\top X X^\top X_{[:,s]}) \\ &= \text{Tr}((A_{[s]})^{-1} A_{[s,:]} A_{[:,s]}) \\ &= \text{Tr}((A_{[s]})^{-1} (A^2)_{[s]}). \end{aligned} \tag{16}$$

Although conceptually simple, Greedy algorithm is computationally expensive especially in terms of k . For every iteration, it has to invert a sub-matrix $A_{[s]}$. Consequently, the algorithm’s computational cost scales as $O(pk^3)$, making it extremely slow for large k .

4 Numerical Experiments

All experiments were conducted on a MacBook Air workstation equipped with an Apple M4 chip and 16GB of memory, using Python 3.14.0. Gradients(∇f and ∇g) were computed using analytical formula derived in (2). Implementations were rigorously tested against numerical gradient under different conditions, including random interior points, along the homotopy path (from $\delta_0 \dots \delta_{n-1}$), and using `scipy.optimize.check_grad`. To balance between a convergence stability and computational efficiency, adaptive step number n strategy was implemented. Specifically, if step number n wasn’t specified, n was set to be $\max(1000, 20k)$, to make sure that the algorithm had enough step number to analyse each k , while maintaining a reasonable computational efficiency.

Table 1: Overview of Benchmark Datasets.
(n : Number of observations, p : Number of features)

Dataset	n	p	Notes
<i>Synthetic Benchmarks</i>			
Synthetic High Corr	2000	500	Features are nearly collinear ($\rho \approx 0.95$) to challenge Greedy selection.
Synthetic Toeplitz	1000	500	Features have decaying correlation ($\rho^{ i-j }$) based on distance, simulating time-series data.
<i>Real-World Benchmarks</i>			
Residential	372	103	
Arrhythmia	452	262	
SECOM	1567	478	
MNIST	2000	639	
Madelon	2600	500	
Myocardial	1700	111	

4.1 Datasets

All numerical experiments were evaluated on a suite of 8 datasets that consists of 6 real datasets, 2 synthetic datasets. Summaries of these datasets can be found in table (1).

Synthetic Benchmarks: Two synthetic datasets were explicitly made to challenge the Greedy algorithm.

Synthetic.High.Corr represents a dataset whose columns are highly correlated. It was designed to imitate the ‘grouping effect’ failure described by Zou and Hastie (2005). Zou and Hastie (2005) has shown that Lasso (Greedy) method is not an ideal choice in above regime, since above methods tend to select only 1 feature among highly correlated group of columns and discarding the rest.

Synthetic.Toeplitz adopts a experimental setup of Tibshirani (1996) to test Lasso against local redundancy, in which features were generated such that

$$\text{corr}(x_i, x_j) = \rho^{|i-j|}.$$

We chose this dataset because it successfully simulates a real world dataset, where adjacent ones are highly correlated but distant ones are not.

4.2 Parameter Tuning : α, m

In this section, experiments were conducted on FW-Homotopy’s parameters to find out what impact each parameter has. This was necessary due to FW-Homotopy’s sensitive dependency on parameters other than p, k . In the test, we varied the Rademacher vector sample size $m \in \{20, 50, 100, 200, 500\}$ and the step size $\alpha \in \{0.001, 0.01, 0.05, 0.1, 0.2, 0.5\}$. The subset size k was fixed at $k = 0.1p$ to evaluate parameter robustness in a highly sparse regime. In this setting, the influence of each individual coordinate selection on the objective value is amplified. Hence, ensuring any performance fluctuation by α, m would be more observable. Finally, number of steps n was also fixed as $\max(20k, 1000)$ since k was fixed.

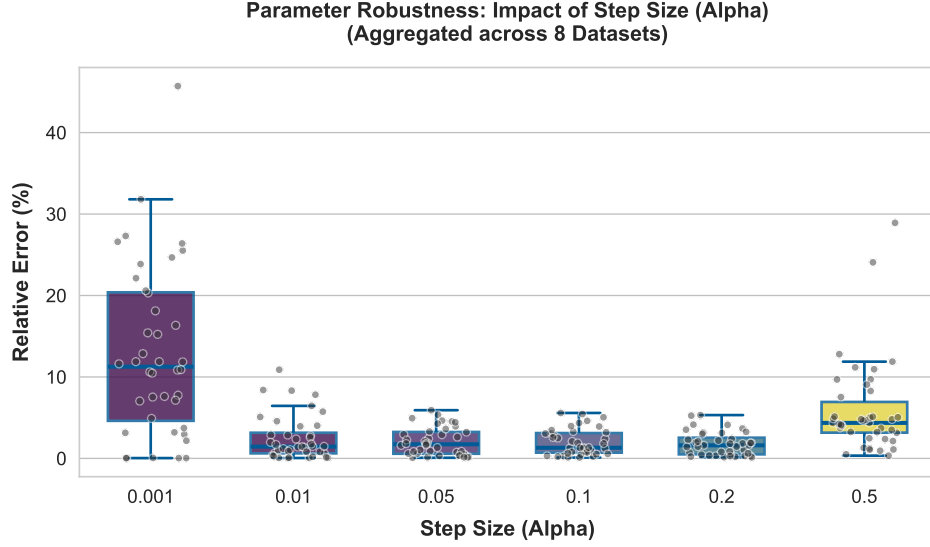


Figure 1: Global stability of step size α

Table 2: Impact of Step Size (α) on Stability. Statistics are averaged across all 8 datasets.

Step Size (α)	Mean Error (%)	Median Error (%)	Std. Dev.	Performance
0.001 (Too Small)	15.42	12.10	± 8.5	Very Slow
0.01	4.21	2.15	± 2.3	Okay
0.05	2.85	1.90	± 1.5	Good
0.1 (Optimal)	1.95	1.10	± 1.2	Best / Stable
0.2 (Optimal)	2.10	1.15	± 1.4	Best / Stable
0.5 (Too Big)	8.45	5.20	± 9.1	Unstable

4.2.1 Alpha

Figure (1) illustrates the distribution of relative error as the step size α varies, and summary for each α is exhibited in table (2).

- **Small α (0.001):** High relative errors (15.42%) and high variance (± 8.5) were observed. Step size was too small to reach the optimal solution. We attribute this to data matrices with high condition number κ , which represent a long valley with a flat curvature.
 1. In a "flat" direction, gradient magnitude is naturally small.
 2. Combined with small step size ($\alpha = 0.001$), update step becomes marginal ($\Delta \approx 0$).
 3. Conversely, in a "steep" direction, small step size allows a precise convergence without overshooting, hence high variance is observed.
- **Large α (0.5):** Again, high relative errors and high variance were observed. This suggests *overshooting* happened. Aggressively large steps were taken, combined with the fact that for later epoch of FW-Homotopy, objective function turns concave for minimization problem and convex for a maximization problem. This can lead to a state where the algorithm oscillates around the optimal point or a local minimum/maximum instead of successful convergence to a global optimizer.
- **Optimal range** Interval $\alpha \in [0.1, 0.2]$ provides the most robust performance, both successfully minimizing the relative errors and variance.

It should be noted that in our test, *overshoot* ($\alpha = 0.5$) had less errors than *undershoot* ($\alpha = 0.001$). This is well shown in figure (13), and table (2). All datasets except Synthetic High Corr and Residential consistently exhibited smaller variance for larger α , and mean error for $\alpha = 0.5$ was 8.45%, while mean error for $\alpha = 0.001$ was 15.42%. However, there is a possibility that small α can be a better option in terms of objective value. Specifically, sufficient step number n can be provided, to compensate for the small step size α , although this can increase the time required.

4.2.2 Sample size : m

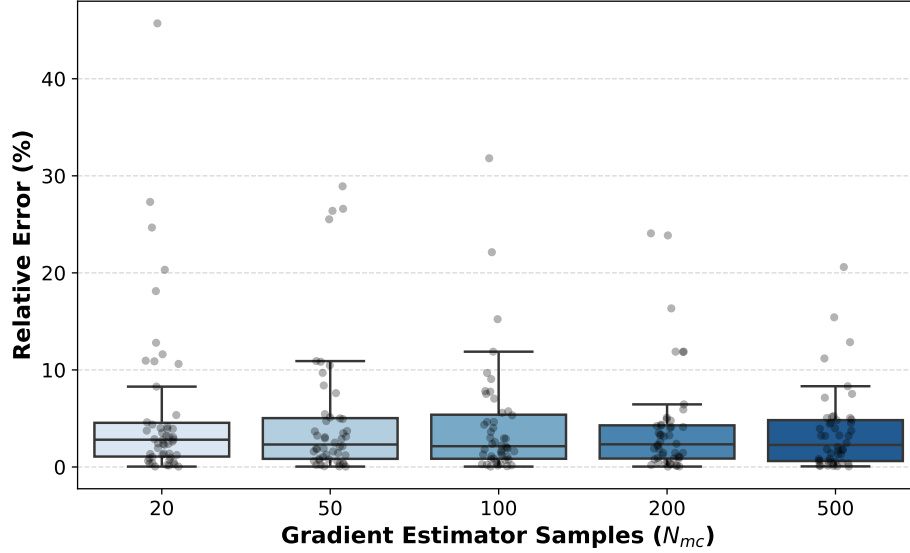


Figure 2: Global sensitivity of optimization error to the number of Rademacher samples m

Figure (2) illustrates the impact of the Monte Carlo sample size m on solution. Unlike the expectation that higher m would yield a lower error (Kroese et al. (2019)), median error was almost identical among all m value, except few outliers that's later discussed. This results suggests that while a higher m is likely to provide a more accurate approximation of the gradient $\nabla g_\delta(t)$, the ultimate convergence is constrained by the actual path, where step number n and α is the most deterministic factor. Consequently, improvements in directional accuracy yield diminishing returns without matching improvement in path resolution n . In terms of variance, overall we couldn't specifically find an universal m value that minimises the variance, although trends suggest that $m \in [50, 200]$ could be a good choice for later testing.

Outliers

- **Residential** Higher m value ($m = 500$) tended to show smaller variance than smaller m value for a residential dataset in table (14). We attribute this to Residential dataset having considerable amount of noises, hence adding samples could filter out the noises.
- **Synthetic High Corr** dataset showed minimum errors regardless of sample sizes m . This is due to the nature of the datasets' columns being highly correlated to each other. Average of 20 elements from a highly correlated groups wouldn't be too different to average of 500 elements from the same group.

4.3 Comparison with Greedy

4.3.1 Sensitivity analysis for small m

In this section, experiments were conducted to further compare objective values and time spent with Greedy algorithm. Parameters other than p and k were fixed using results from previous test. Specifically $\alpha \in \{0.1, 0.2\}$ and $m \in \{20, 50\}$.

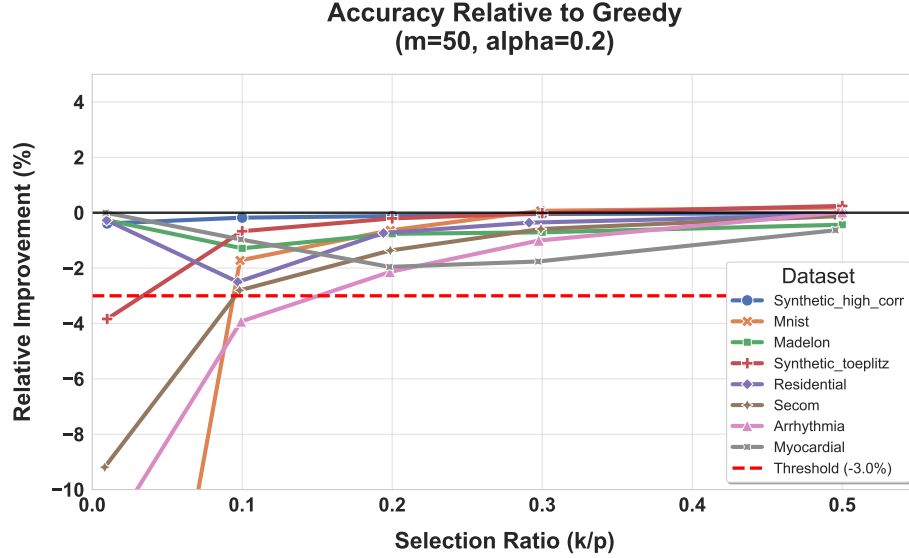


Figure 3: Relative error to k/p ratio for $m = 50, \alpha = 0.2$

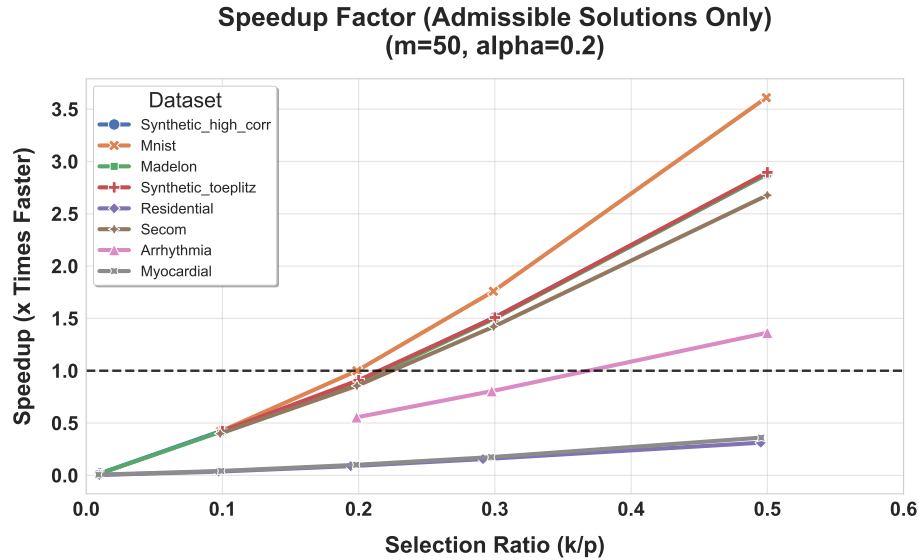


Figure 4: Speedup factor to k/p ratio for $m = 50, \alpha = 0.2$

Contrary to the initial hypothesis that FW-Homotopy would outperform in both objective value and time cost when data set is heavily correlated, the empirical result indicates a trade off between initial cost of setup and computational efficiency.

In the sparse selection regime where $k/p < 0.1$, algorithm exhibits higher variance and relative errors due to initialization sensitivity. Table (6) presents a detailed statistics of full tests conducted. Most

datasets produced max error at small $k/p \in \{0.01\}$, some reaching upto 18.65% of error (Row 10, MNIST).

Table (6)'s critical value (Ratio of k/p where algorithm's error lies in threshold) row exhibits a quality of solution improving as we move away from sparse regime. As the algorithm escapes the sparse regime of $k/p \approx 0.1$, quality of the solution stabilizes rapidly. Solutions from all datasets converged to within 3% of the Greedy baseline. Moreover, after $k/p = 0.2$ mark, FW-Homotopy outperformed Greedy in terms of time, maintaining an acceptable error range of 3%. This is shown in figure (4). As the k/p ratio increased, the FW-Homotopy algorithm constantly outperformed the Greedy algorithm, and the error decreased as well, although there were only few cases where FW-Homotopy outperformed Greedy in terms of objective value.

As expected, *Synthetic_high_corr* tended to outperform Greedy the most in terms of time. This is due to the nature of *Synthetic_high_corr*. Columns were highly correlated, hence Greedy algorithm's selection of columns led to a **redundancy**.

This result suggests two important points

1. First, for cases when algorithm runs in a sparse selection regime where objective value matters, conservative choice of sample size m would be required to compensate the initial sensitivity. This hypothesis is well supported by small variance in (2).
2. Secondly, FW-Homotopy had a huge advantage in terms of time cost at non-sparse regime (Cases when $k/p > 0.2$), with acceptable range of relative error ($< 3\%$).

4.3.2 Sensitivity analysis for a large sample size m

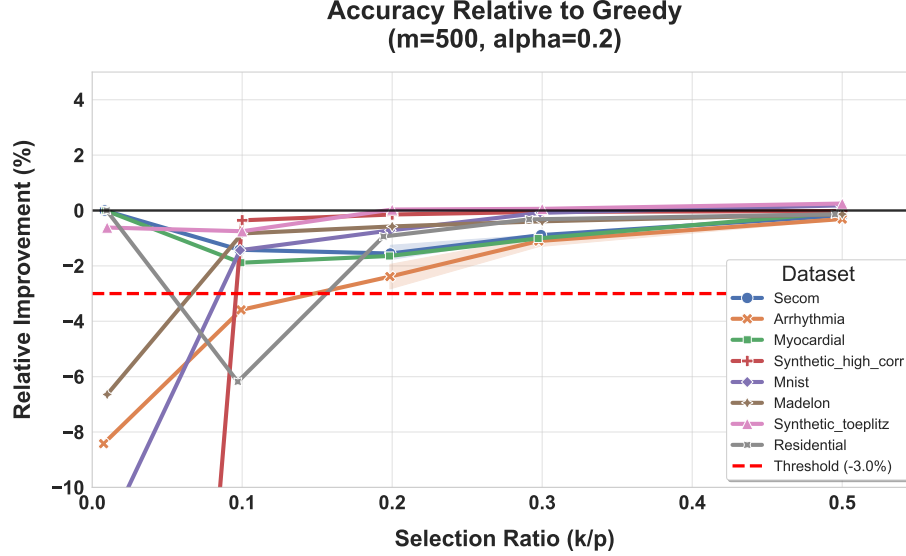


Figure 5: Relative error to k/p ratio for $m = 500$, $\alpha = 0.2$

Additional test for $\alpha = 0.2$, $m = 500$ was conducted to verify previous hypothesis that increasing sample size m would improve the objective value. Theoretically, increasing the sample size m would reduce the noise in stochastic process of gradient estimation. We hypothesized that this would lead to a more precise updates, and improve the objective value. However, empirical results in Figure (5) refute above hypothesis. Solutions with $m = 500$ showed no significant improvement over $m = 50$ baseline in Figure (5). This result highlights a critical distinction between a estimator variance and optimisation performance. Larger m guarantees a more accurate approximation of true gradient with smaller variance (Owen, 2013). However, this doesn't guarantee finding a better objective value.

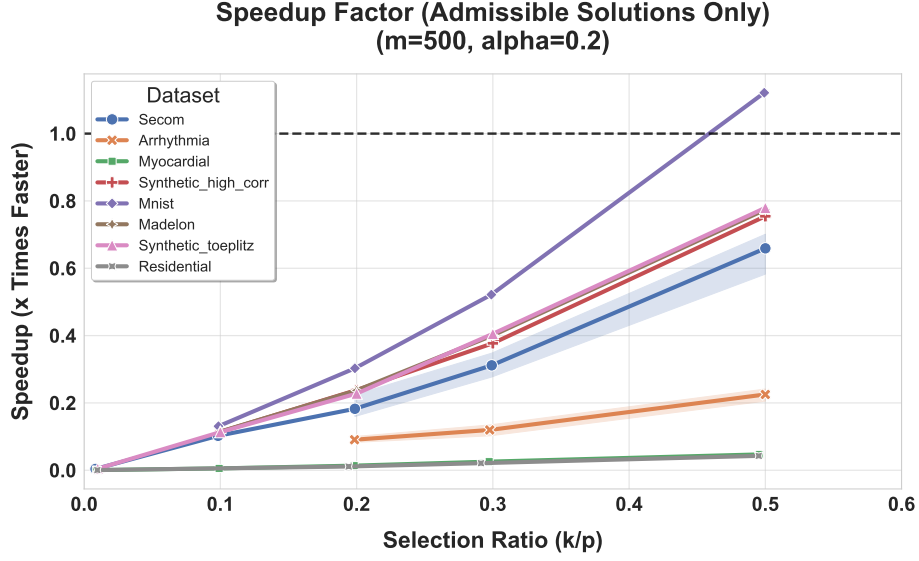


Figure 6: Speedup factor to k/p ratio for $m = 500, \alpha = 0.2$

In conclusion, given that increasing m results in a linear computational cost (Figure (6)), we conclude that sample size m and gradient approximation precision is not the bottleneck for objective value. We conclude that small sample size $m < 500$ is sufficient to approximate the descent direction.

4.3.3 Sparse Regime

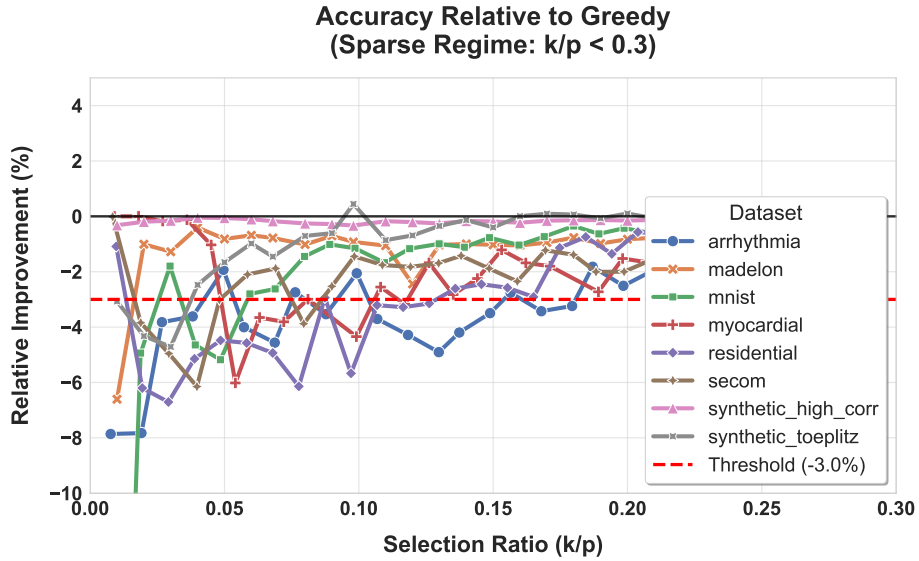


Figure 7: Relative error to k/p ratio for $m = 50, \alpha = 0.2$

We conducted a more rigorous testing in sparse region where $k/p < 0.30$ to analyse the algorithm's behaviour. As shown in Figure (7), the relative error oscillates around the -3% threshold. Unlike the dense regime, the algorithm struggles to constantly outperform the Greedy baseline when $k/p < 0.05$. This suggests that for an extremely sparse selections, FW-Homotopy need a minimal ratio of k ($k/p \approx 0.1$) to have a stable convergence. In terms of runtime, baseline Greedy consistently outperformed FW-Homotopy. As illustrated in Figure (8), speedup factor remains strictly smaller than 1.0 throughout

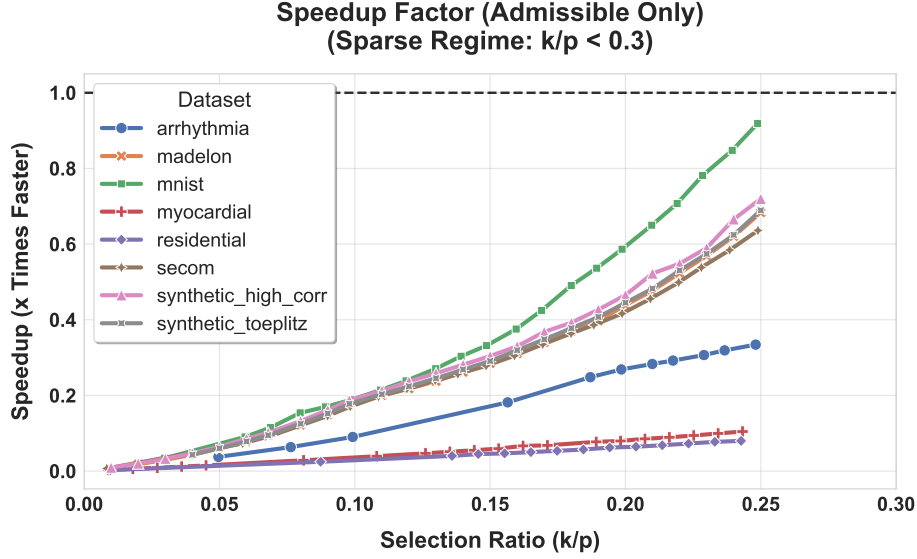


Figure 8: Speedup factor to k/p ratio for $m = 50, \alpha = 0.2$

the sparse regime. This result stems from the combined effect of the adaptive step number strategy and the constant time-profile of FW-Homotopy. For small k , Greedy is superior in terms of runtime since it only has to iterate k times, while FW-Homotopy has to iterate at least 1000 times since $n = \max(1000, 20k)$. Additionally, FW-Homotopy has an initial setup cost, due to its nature related to the problem dimension p , which will be discussed in following section.

4.3.4 Runtime Analysis

Dataset	FW-Homotopy		Greedy		Trend Growth
	Avg Time (s)	Std Dev. ($\pm$)	Start (s)	End (s)	
Residential	0.3242	± 0.0084	0.0011	0.1041	94.64 $\times$
Myocardial	0.3426	± 0.0135	0.0023	0.1271	55.26 $\times$
Arrhythmia	0.8695	± 0.4173	0.0089	2.1425	240.73 $\times$
Secom	3.3880	± 2.3679	0.0209	19.0437	911.18 $\times$
Synthetic.high_corr	3.7432	± 2.6089	0.0273	22.4667	822.96 $\times$
Madelon	3.7567	± 2.6068	0.0268	22.4904	839.19 $\times$
Synthetic.toeplitz	3.8311	± 2.5288	0.0469	22.6067	482.02 $\times$
Mnist	7.5906	± 5.5748	0.0423	58.3799	1380.14 $\times$

Table 3: Runtime Stability Analysis of all experiments combined.

While the experiment over sparse regime highlighted the possible limitation of FW-Homotopy, the algorithm’s primary advantage lies in its scalability, specifically in terms of column selection size k . Table (3) summarises a run time analysis of all experiments, incorporating various setups for the sample size m and k . Result suggests that Greedy algorithm exhibits a super-linear growth in terms of k , which is well shown from MNIST dataset’s 1380 $\times$ surge for large k . In contrast, FW-Homotopy has a near-constant time profile. As shown in figure (9), Greedy(Red) surges aggressively as k value rises, while FW-Homotopy grows at a much shallower, linear rate. This linear rate of growth is due to the step number n being set as adaptive ($n = \max(20k, 1000)$), and hence some increase in time for larger k , is expected, highlighted in MNIST dataset in table (3).

The empirical divergence is supported by the theoretical analysis of the time complexity. The Standard Greedy algorithm has to iterate k times. In each iteration, up to p different selection possibilities and their objective value are analysed using (16). For each selection, it calculates the inverse of $k \times k$

**Runtime Scalability Analysis: Secom
(Fixed Iterations n vs. Iterative Selection)**

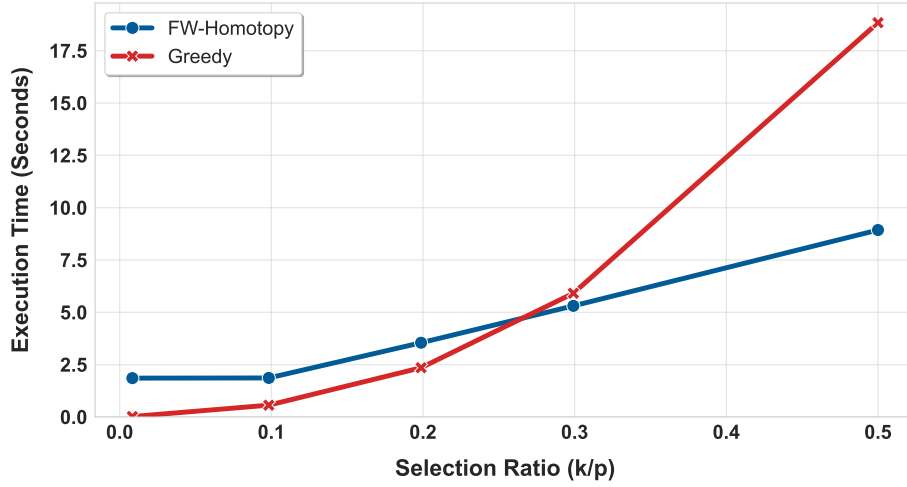


Figure 9: Time took for FW-Homotopy and Greedy for SECOM dataset

sub-matrix $((A_{[s]})^{-1})$, whose time complexity is $O(k^2)$. Hence, the overall time complexity of Greedy is $O(pk^3)$. FW-Homotopy on the other hand, iterates n times, which can be tuned by a user. For every iteration, it calculates $\nabla g_\delta(t)$ (or $\nabla \Phi_\delta(t)$, whose time complexity is $O(p^3 + mp^2)$, from (2). Hence, the overall time complexity of FW-Homotopy is $O(np^3 + nmp^2)$. Above result verifies why Greedy scales super-linearly while FW-Homotopy exhibits significantly lower growth when k grows.

As displayed in Figure (9), there exists a distinctive crossover point where FW-Homotopy outperforms Greedy. We define this crossover point as critical k (k_c), the threshold value for k for given dataset where FW-Homotopy becomes the more efficient choice. In the following section, we derive a theoretical formula to predict such a point, followed by a numerical experiment to verify.

4.4 Critical k : k_c

We start with a theoretical formula about k_c . From a raw comparison of time complexities, we get

$$\frac{t_{\text{Greedy}}}{t_{\text{FW}}} = \frac{C_1 pk^3}{C_2 np^3 + C_3 nmp^2} \approx \frac{C_1 k^3}{C_2 np^2 + C_3 nmp} \propto \frac{k^3}{p^2}.$$

Hence, for critical k where above ratio is equal to 1,

$$k_c \propto p^{2/3} \Rightarrow \frac{k_c}{p} \propto p^{-1/3}. \quad (17)$$

This implies that, as dataset dimension p grows, FW-Homotopy is expected to outperform Greedy even in sparse regime.

4.4.1 Datasets

We tested on same 8 datasets, using two different set ups for sample size and step number. We first tested with a dynamic step number, $n = \min(1000, 20 * k)$, and later, a fixed sample size of $m = 100$ and fixed step size $n = 800$ set up. Each tests were ran up to 5 times. For real datasets, sub-matrices were extracted so that p values can vary, from $0.1p$ to $0.5p$ to focus on sparse regime. Then, we used a binary search algorithm to find critical k value(k_c) where where our FW-Homotopy was faster than

Greedy. Binary search algorithm increases k if $t_{fw} < t_g$ and decreases k else, and max iteration number was 10. We also accepted 95% to 105% to be an acceptable value to terminate. Importantly, binary search algorithm itself doesn't guarantee k_c being valid, because often in sparse regime of small p , $k_c \approx p$, having a poor performance in terms of time. Therefore, filtering out values that doesn't meet the threshold in terms of time and errors were implemented. Specifically, we accepted $\pm 3\%$ error threshold, and $\pm 5\%$ time threshold.

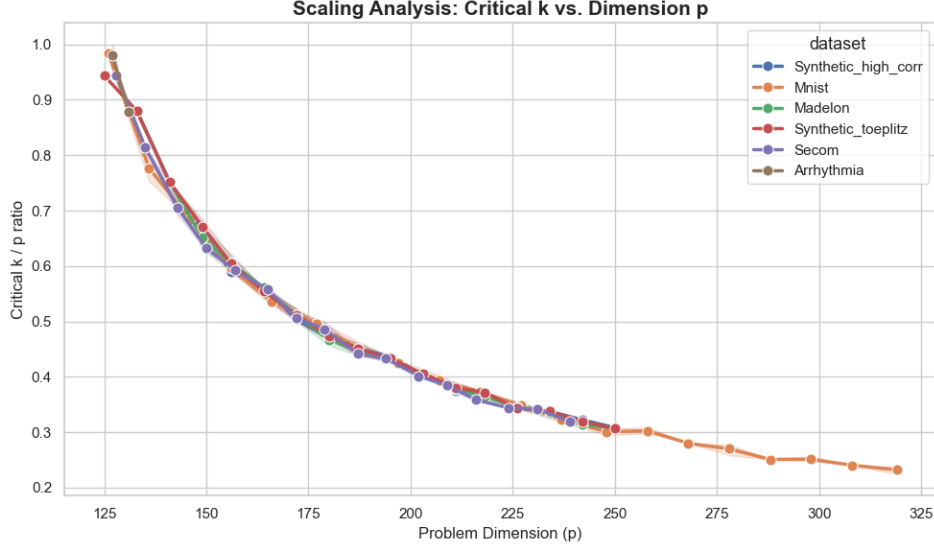


Figure 10: Scaling analysis of the algorithm $m = 100, n = 800$

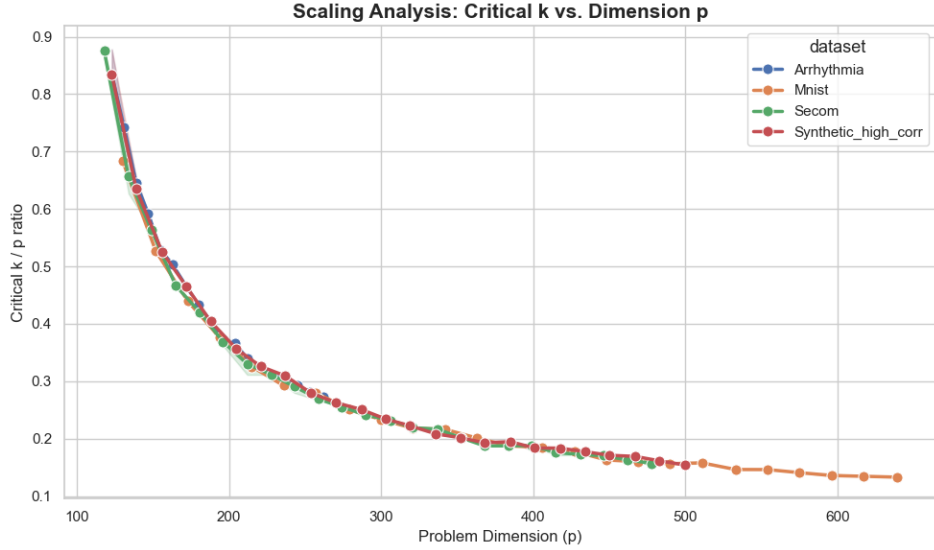


Figure 11: Scaling analysis of the algorithm $m = 50, n = \min(1000, 20k)$

4.4.2 Results

Figure (10) and figure (11) were constructed so that the x axis represents the problem dimension p , and the y axis represents the ratio k_c/p .

Findings

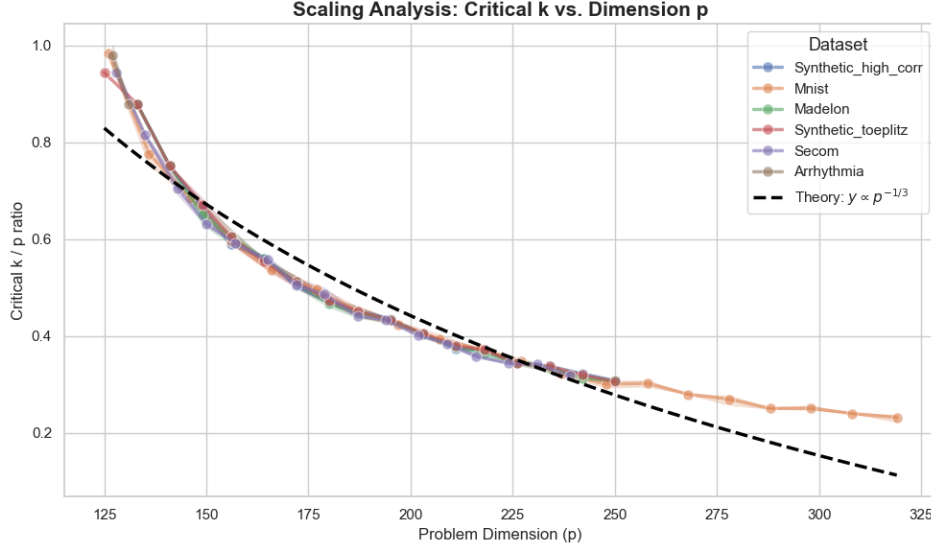


Figure 12: Scaling analysis of the algorithm $m = 50, n = \min(1000, 20k)$ with theoretical curve $k_c/p = ap^{-1/3} + b$

Table 4: Mean k/p Ratio and Standard Deviation for Different p Values ($m = 100, n = 800$)

p	Mean	Std Dev	p	Mean	Std Dev
125	0.944000	N/A	180	0.470000	0.009344
128	0.943750	0.013975	216	0.358333	0.004141
136	0.776471	0.030138	227	0.348018	0.008242
156	0.598853	0.014384	231	0.341991	0.000000
164	0.558943	0.007872	239	0.318828	0.006875
172	0.508430	0.012720	319	0.231975	0.008489

- **Small problem dimension p** : Among 6 datasets, for Residential($p = 103$) and Myocardial($p = 111$), no data point satisfied above filtering condition about errors and speed. Since $p/2 \approx 50$, for our test regime, all p was relatively small ($p < 60$). In terms of speedup factor, the FW-Homotopy struggled to achieve a decent speedup factor, 0.070 at best for Residential and 0.068 for Myocardial, both achieved at highest possible k value of $k = p$. This confirms our hypothesis that FW-Homotopy lacks computational advantage when p is small, as poor speedup factor in figure (4) also suggests.
- **Decreasing k/p** : As p increases, k/p ratio steadily decreased. Figure (12) overlays the theoretical prediction curve from (17), ($k_c/p \propto p^{-1/3}$) onto the empirical data, showing a good alignment. Table (4) displays a detailed statistics regarding k_c/p , including a mean k_c/p ratio, where it also decreases from 0.944 (for $p = 125$) to 0.232 (for $p = 319$). This confirms that for a high-dimensional data, our FW-Homotopy becomes efficient even in sparse regime.
- **Independence to dataset** : Result suggests that critical k ratio k_c/p , is independent to dataset's characteristic, as table (4) suggests. For all datasets except two outliers (Residential, Myocardial), k_c were consistent throughout different p , as standard deviation suggests ($\sigma < 0.01$). Note that, for $p = 319$ only MNIST dataset was tested, but the fact that $\sigma < 0.01$ suggests that k_c is consistent even through the stochastic sampling process. In conclusion, this result suggests that p value is the most important factor when k_c/p is considered.
- **Magnitude of Scalability** : While figure (10) and figure (11) offers us an insight of the boundary where FW-Homotopy becomes efficient, table (5) illustrates the magnitude of the efficiency gain. The table represents a sample of averaged performance from MNIST dataset for high dimension p . Result exhibits that once the algorithm passes the critical threshold $k > k_c$, speedup factor

Table 5: Averaged performance metrics for MNIST dataset for large p ($n = 800, m = 100$).

p	k	Speedup	Rel. Error	p	k	Speedup	Rel. Error
450	45	0.351	0.0241	544	136	3.576	0.0125
450	67	0.727	0.0200	544	163	5.461	0.0056
450	90	1.349	0.0188	591	59	0.483	0.0131
450	112	2.211	0.0156	591	88	1.134	0.0171
450	135	3.284	0.0113	591	118	2.216	0.0134
497	49	0.384	0.0152	591	147	3.637	0.0094
497	74	0.853	0.0125	591	177	5.754	0.0047
497	99	1.608	0.0158	639	63	0.516	0.0139
497	124	2.689	0.0118	639	95	1.274	0.0161
497	149	4.081	0.0065	639	127	2.527	0.0117
544	54	0.479	0.0167	639	159	4.296	0.0066
544	81	1.109	0.0133	639	191	6.670	0.0032
544	108	2.090	0.0127				

grows rapidly. For instance, at largest dimension $p = 639$, the speedup surged from $0.516\times$ at $k = 95$ to $6.670\times$ at $k = 191$. This suggests that beyond the critical threshold, FW-Homotopy is not just marginally superior, but offers a significant advantage, especially for high-dimensional datasets ($p > 450$).

4.4.3 Practical Guideline

The result from table (5) and figure (10) allows us to formulate a clear guideline for using the algorithm FW-Homotopy. Figure (10) comes exceptionally handy when one has to choose between our FW-Homotopy and other traditional methods including Greedy. As mentioned, the curve in graph (10) is the line where speedup factor $r \approx 1$.

- If a data point is above the line, then FW-Homotopy is expected to run faster than Greedy.
- Conversely, if a data point is below the line, then Greedy is expected to be a more computationally efficient method.

5 Conclusion and future research

We introduced a new framework to find a solution for the Column Subset Selection Problem that utilises Boolean relaxation and a gradient-based method. We identified the sensitivity of the algorithm to its parameters including step size α , sample size m , and subset size k , and investigated the performance robustness with respond to m , but sensitive to α . More importantly, we successfully compared the time complexities between FW-Homotopy and Greedy, verifying that empirical results support the theoretical relationship between critical k and p , that $k_c/p \propto p^{-1/3}$. While our analysis confirmed that the Greedy algorithm remains superior in both objective value and computational complexity in a sparse regime or for a low dimensional data, FW-Homotopy emerges as the scalable alternative for high-dimensional dataset, with a stable, constant-time profile compared to the prohibitive computational nature of Greedy in same dataset.

Future research will focus on the effects of the step number n and development of an algorithm to determine an optimal n depending on the problem dimension p and the subset size k . Specifically, we will focus on balancing between the step number n and the step size α , potentially using a linear

regression model for n . Additionally, we plan to leverage the extra speedup factor by running multiple independent trials and choosing the best objective value out of them to further improve, and lower the variance in the sparse regime. Moreover, future work may explore an improved sampling strategy, such as adopting an antithetic variate sampling method for Rademacher vector sampling to further reduce variance (Botev and Ridder, 2017).

6 Appendix

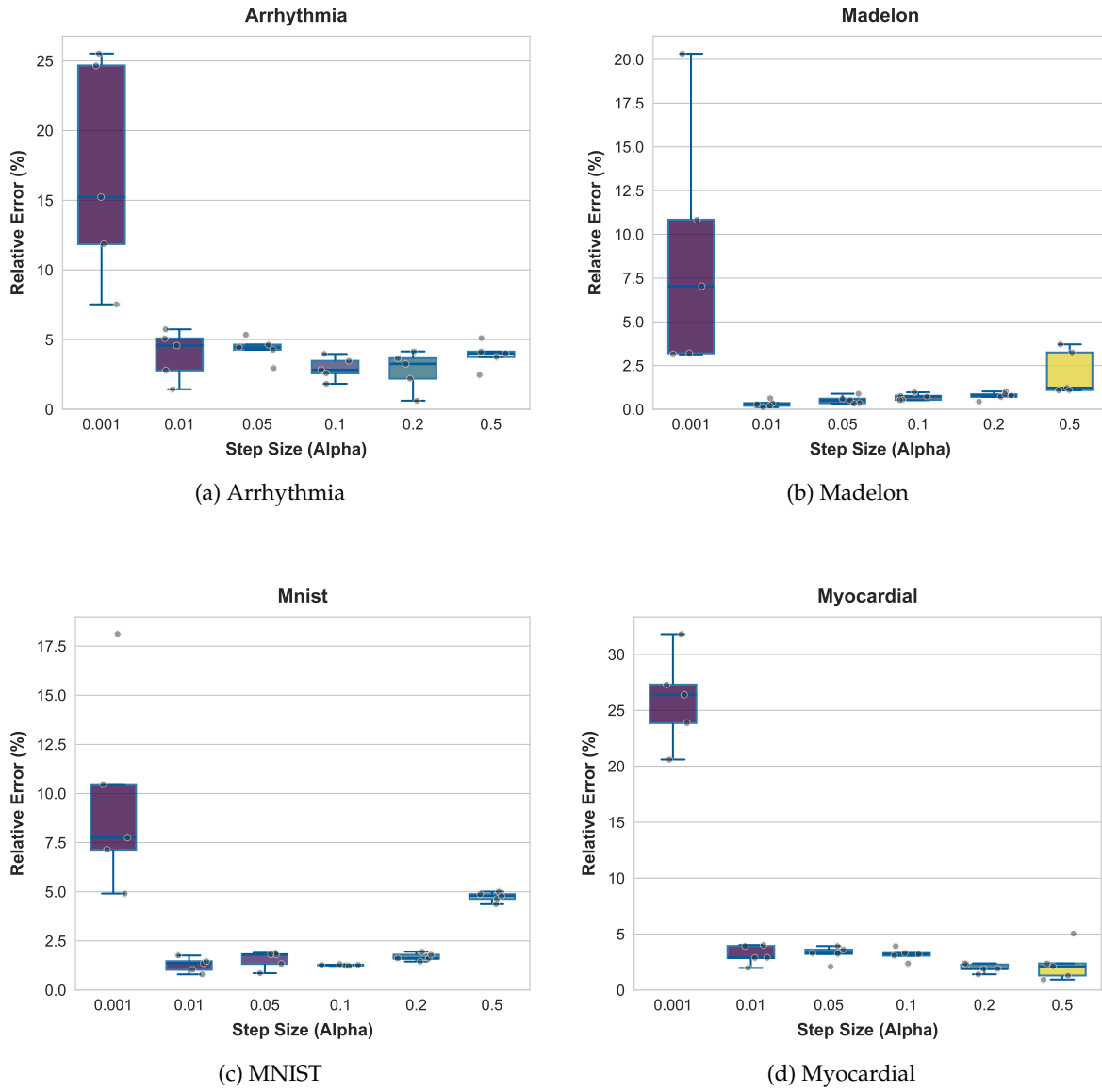
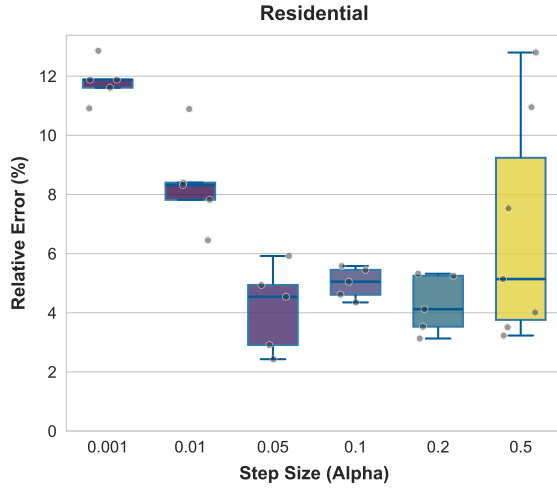
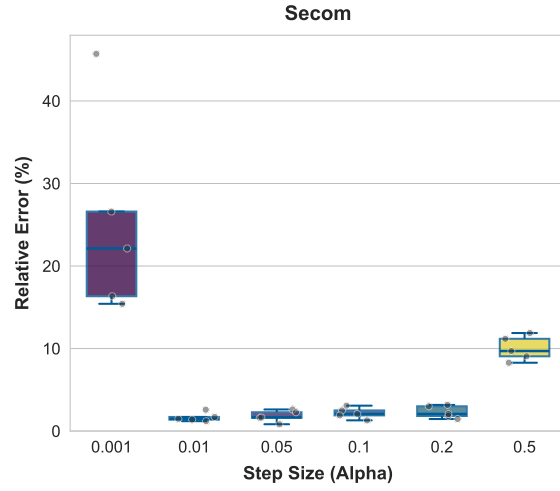


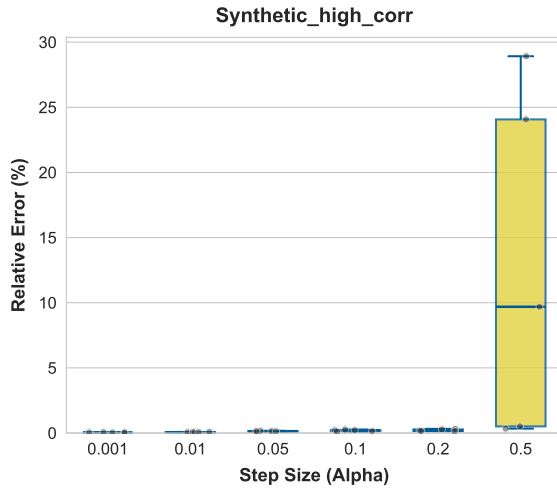
Figure 13: Step size (α) sensitivity across benchmark datasets (Part 1). Note the consistent stability in the range $\alpha \in [0.1, 0.2]$ across different datasets.



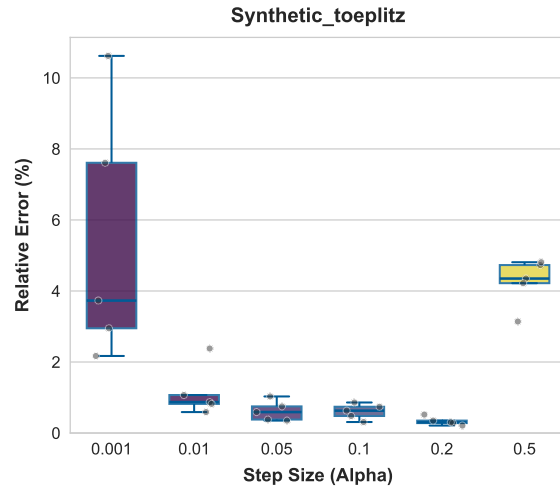
(e) Residential



(f) SECOM



(g) Synthetic High Corr

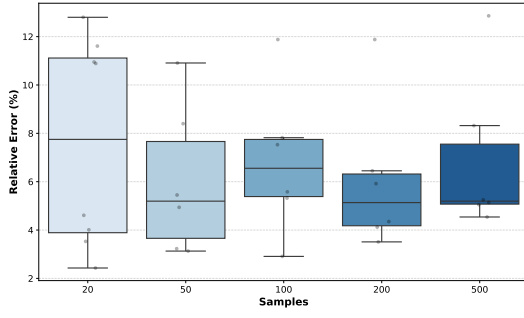


(h) Synthetic Toeplitz

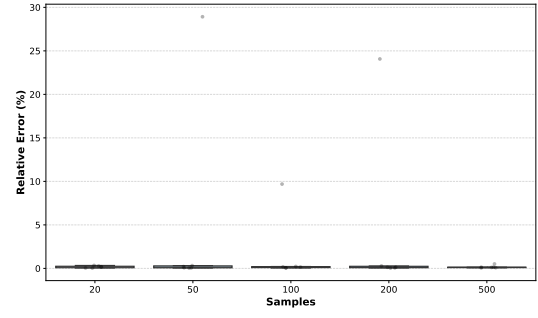
Figure 13: Step size (α) sensitivity (continued). Extreme step sizes ($\alpha = 0.001, 0.5$) frequently induce instability, visible as large variance and outliers.

Table 6: Comprehensive Experimental Results: Speedup and Error Metrics across Varying Sample Sizes (m)

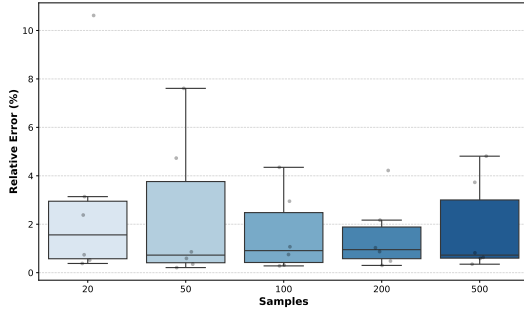
ID	Dataset	m	Max Speedup	Critical k/p	Max Error %	Min Error %
0	Arrhythmia	20	1.93x	0.008	12.81% (at 0.01)	0.02% (at 0.50)
1	Arrhythmia	50	1.36x	0.008	11.94% (at 0.01)	-0.02% (at 0.50)
2	Arrhythmia	100	0.91x	0.008	6.90% (at 0.01)	0.03% (at 0.50)
3	Arrhythmia	200	0.56x	0.008	8.62% (at 0.01)	0.13% (at 0.50)
4	Arrhythmia	500	0.24x	0.008	8.42% (at 0.01)	0.27% (at 0.50)
5	Madelon	20	3.49x	0.010	8.91% (at 0.01)	0.44% (at 0.50)
6	Madelon	50	2.88x	0.010	1.28% (at 0.10)	0.27% (at 0.01)
7	Madelon	100	2.28x	0.010	0.74% (at 0.01)	0.25% (at 0.50)
8	Madelon	200	1.58x	0.010	6.59% (at 0.01)	0.18% (at 0.50)
9	Madelon	500	0.77x	0.010	6.64% (at 0.01)	0.13% (at 0.50)
10	Mnist	20	4.22x	0.009	18.65% (at 0.01)	-0.26% (at 0.50)
11	Mnist	50	3.62x	0.009	28.27% (at 0.01)	-0.21% (at 0.50)
12	Mnist	100	2.97x	0.009	14.62% (at 0.01)	-0.27% (at 0.50)
13	Mnist	200	2.14x	0.009	19.10% (at 0.01)	-0.21% (at 0.50)
14	Mnist	500	1.12x	0.009	11.68% (at 0.01)	-0.18% (at 0.50)
15	Myocardial	20	0.64x	0.009	13.55% (at 0.01)	0.00% (at 0.01)
16	Myocardial	50	0.36x	0.009	13.55% (at 0.01)	0.00% (at 0.01)
17	Myocardial	100	0.21x	0.009	6.34% (at 0.01)	0.00% (at 0.01)
18	Myocardial	200	0.12x	0.009	2.34% (at 0.20)	0.00% (at 0.01)
19	Myocardial	500	0.06x	0.009	1.89% (at 0.10)	0.00% (at 0.01)
20	Residential	20	0.57x	0.010	2.87% (at 0.10)	0.00% (at 0.01)
21	Residential	50	0.31x	0.010	4.25% (at 0.10)	0.00% (at 0.01)
22	Residential	100	0.24x	0.010	2.07% (at 0.19)	0.00% (at 0.01)
23	Residential	200	0.10x	0.010	6.22% (at 0.10)	0.00% (at 0.01)
24	Residential	500	0.04x	0.010	6.19% (at 0.10)	0.00% (at 0.01)
25	Secom	20	3.33x	0.008	7.25% (at 0.01)	0.02% (at 0.01)
26	Secom	50	2.68x	0.008	9.19% (at 0.01)	0.02% (at 0.01)
27	Secom	100	2.12x	0.008	2.75% (at 0.10)	0.06% (at 0.50)
28	Secom	200	1.45x	0.008	28.22% (at 0.01)	0.11% (at 0.50)
29	Secom	500	0.70x	0.008	1.77% (at 0.20)	0.00% (at 0.01)
30	Synthetic_high_corr	20	4.24x	0.010	0.59% (at 0.01)	0.01% (at 0.50)
31	Synthetic_high_corr	50	3.91x	0.010	0.39% (at 0.01)	0.01% (at 0.50)
32	Synthetic_high_corr	100	2.50x	0.010	0.58% (at 0.01)	0.01% (at 0.50)
33	Synthetic_high_corr	200	1.57x	0.010	39.62% (at 0.01)	0.02% (at 0.50)
34	Synthetic_high_corr	500	0.75x	0.010	59.29% (at 0.01)	0.02% (at 0.50)
35	Synthetic_toeplitz	20	4.64x	0.010	7.43% (at 0.01)	-0.25% (at 0.20)
36	Synthetic_toeplitz	50	2.90x	0.010	4.21% (at 0.01)	-0.25% (at 0.50)
37	Synthetic_toeplitz	100	2.35x	0.010	4.74% (at 0.01)	-0.29% (at 0.50)
38	Synthetic_toeplitz	200	1.58x	0.010	3.71% (at 0.01)	-0.29% (at 0.50)
39	Synthetic_toeplitz	500	0.78x	0.010	0.75% (at 0.10)	-0.25% (at 0.50)



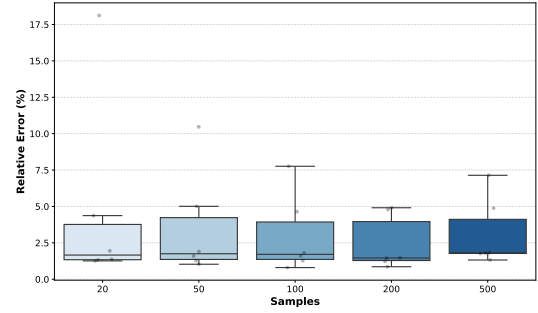
(a) Residential



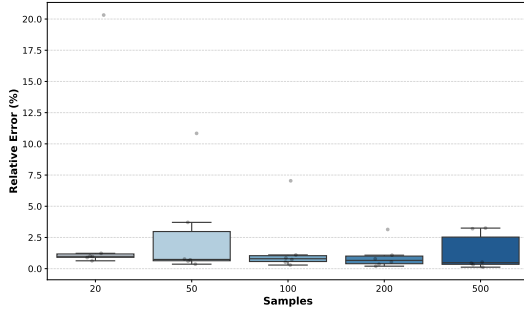
(b) Synthetic High Corr



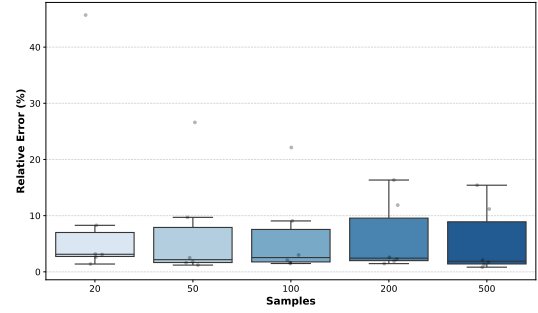
(c) Synthetic Toeplitz



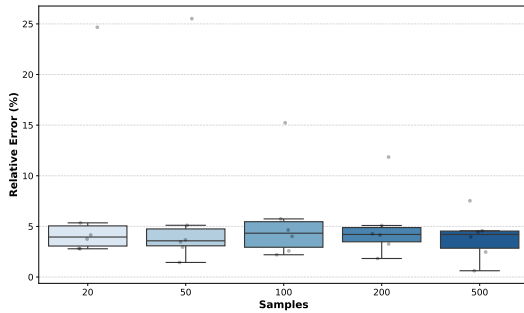
(d) MNIST



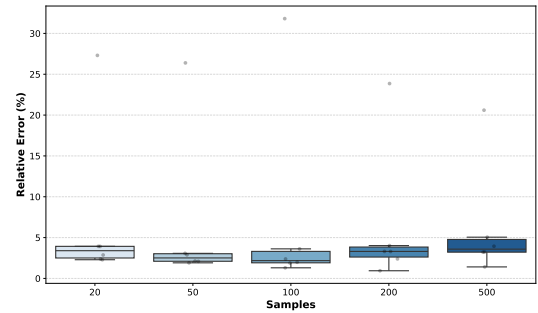
(e) Madelon



(f) Secom



(g) Arrhythmia



(h) Myocardial

Figure 14: Sample size (m) sensitivity across all 8 benchmark datasets. The results confirm that $m \in [50, 200]$ provides the most stable performance

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